

PHIL 470 - Logical Theory II

Incompleteness and Undecidability

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Semester:	Spring 2026
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Office Hours:	TBA
Class Times:	TuTh 2:00pm - 3:15pm
Class Location:	TWS 0201

Course Description

This course will focus on Kurt Gödel's first and second incompleteness theorems. The first incompleteness theorem states, roughly, that every logical system that is sufficiently expressive and free from contradictions is incomplete in the sense that there are always statements such that neither the statement nor its negation can be proved. The second incompleteness theorem states that sufficiently expressive arithmetic theories cannot prove their own consistency. We will prove the 1st and 2nd Incompleteness Theorems and survey their technical and philosophical repercussions.

The primary goal of the course is to introduce the technical and philosophical topics that arise when proving Gödel's Incompleteness Theorems. Topics to be covered include: formal models of computation (especially elementary recursion theory); the Church-Turing Thesis; Gödel's 1st and 2nd incompleteness theorems and their repercussions; Tarski's proof of the undefinability of truth; Undecidability of the Halting Problem; provability logic - Kripke soundness and completeness (time permitting), absolute provability and The Knower Paradox (time permitting); and non-standard models of arithmetic (time permitting).

Prerequisites: PHIL370 (or equivalent logic course), or permission from the instructor.

Reading

The readings for the course will be made available on the course website. Readings will be taken from the following texts:

1. *Incompleteness and Undecidability*. This text is compiled from the Open Logic Project: (openlogicproject.org).
2. J. Avigad, *Mathematical Logic and Computation*, Cambridge University Press, 2023.
3. J. D. Hamkins, *Lectures on the Philosophy of Mathematics*, the MIT press, 2021.

4. P. Smith, *An Introduction to Gödel's Theorems*, Second Edition, Cambridge, 2020. <https://www.logicmatters.net/resources/pdfs/godelbook/GodelBookLM.pdf>.
5. P. Raatikainen, "Gödel's Incompleteness Theorems", The Stanford Encyclopedia of Philosophy, <https://plato.stanford.edu/entries/goedel-incompleteness/>.

The following texts are recommended for additional reading:

- G. Boolos, J. Burgess, and R. Jeffrey, *Computability and Logic*, Cambridge, 2007.
- I. Chiswell and W. Hodges, *Mathematical Logic*, Oxford University Press, 2007.
- H.-D. Ebbinghaus, J. Flum, and W. Thomas, *Mathematical Logic*, Springer, 1995.
- H. Enderton, *A Mathematical Introduction to Logic*, Academic Press, 2nd Edition, 2001.
- M. Fitting, *Incompleteness in the Land of Sets*, Kings College Press, 2007.
- T. Franzen, *Gödel's Theorem: An Incomplete Guide to its Use and Abuse*, A K Peters, 2005.
- R. Smullyan, *Gödel's Incompleteness Theorems*, Oxford University Press, 1992.

Grading Policy

The course requirements are:

- **Participation (20%)**: This is a lecture course and we have a lot of material to cover, so you will easily fall behind if you miss lectures. You are responsible for learning any material covered in lectures that you miss. **Please email me if you are going to miss class.** Your participation grade will be based on a weekly question posted on the discussion board:

Post one question about this week's material on the ELMS discussion board. Your question might ask for clarification of an idea or theorem, explanation of a step in a proof, or help with a problem set question. If you genuinely have no questions this week, you may submit "No question this week", but you may do this **at most 3 times during the semester**. A fourth "no question" submission will receive a 0. Questions will be graded as follows: 1 for any genuine question (or permitted "no question"), 0 for no submission or a question unrelated to course content. Note that you can use LaTeX for notation: For instance, type

$\backslash(\backslash\text{forall } x \backslash\text{forall } y \ x+\mathbf{S}(y) = \mathbf{S}(x+y)\backslash)$
to get $\forall x \forall y \ x + \mathbf{S}(y) = \mathbf{S}(x + y)$.

- **Problem Sets (40%)**: There are three components of the Problem Set grade:
 - **Basic Problems (50% of Problem Set score)**: Foundational problems reinforcing core definitions and ideas. You may work at your own pace with unlimited opportunities to revise based on my feedback. I will grade partially completed submissions. Problem sets will be released throughout the semester with suggested due dates. All must be submitted by **May 8**.

- **Advanced Problems (30% of Problem Set score):** 3–4 problem sets assigned throughout the semester, each due approximately 10 days after the problems are released. Attempt as many problems as you can, but you need not solve every one. The goal is clear presentation, not just correctness.
- **Proof Portfolio (20% of Problem Set score):** At the end of the semester, select 3–4 proofs. Revise them into polished final versions demonstrating clear, complete mathematical writing. Due during exam week.

Collaboration: You may discuss problems with others in the class and with me, but all write-ups must be entirely your own.

- **Check-up Exams (20%):** Four short in-class exams throughout the semester, approximately 20-30 minutes each.
 - Background: sets, functions, induction, first-order logic (Week 4)
 - Formal Arithmetic and primitive recursive functions (Week 7)
 - Representability (Week 10)
 - Fixed-point theorem and incompleteness (Week 13)

If you are unsatisfied with your score, you may retake a check-up exam once during office hours within one week of receiving your grade. The retake replaces your original score.

- **Final exam (20%).** The final exam will be an in-class exam given during exam week. A review sheet containing the material that will be covered on the exam will be provided a week or two in advance of the exam.

Tentative Syllabus

Below is a tentative schedule for the semester (consult the course website for more details).

1. Introduction to Incompleteness (1 week)
2. Background: Sets, functions, relations, enumerations, mathematical induction (1 week)
3. Background: first order logic, natural deduction (1 week)
4. Formal Arithmetic (1 week)
5. Computability, Primitive Recursive Functions, Recursive Functions (2 weeks)
6. Arithmetization of syntax (Gödel numbering) (1 week)
7. Representability in $\mathcal{Q}$ (2 weeks)
8. The Diagonal Lemma and the First Incompleteness Theorem (1-2 week)
9. The Second Incompleteness Theorem (1 week)
10. Derivability conditions and Löb's Theorem (1-2 weeks)

Time permitting we will cover some of the following topics (depending on student interest).

1. Gödel's Disjunction, Absolute Provability and Epistemic Arithmetic, Montague and Kaplan's Knower Paradox
2. Models of arithmetic
3. Provability Logic
4. Second-order arithmetic

Important Dates

- Spring Break: March 15 – 22, No class on Tuesday, March 17 and Thursday, March 19.
- In-person lectures canceled, away at a conference: Tuesday, March 31 and Thursday, April 2.
- Last Class: Thursday, May 7.
- Exam Week: May 11 - 18 (the date of the exam will be announced later in the semester when it is scheduled).

Learning Outcomes

Students who successfully complete this course will be prepared to:

- Define the logical systems and main concepts (including mathematical models of computation) used in the proof of the incompleteness theorems
- Reproduce an outline of the proof of Gödel's Incompleteness Theorems and related results
- Explain the philosophical implications of Gödel's Theorems and related results
- Discuss the Church-Turing thesis and the undecidability phenomenon

Course Policies

The course is subject to the various policies found here:

<http://www.ugst.umd.edu/courserelatedpolicies.html>

In addition, there is the further e-free policy: please minimize the use of laptops, phones, or other mobile devices during class.